

M08 Protected Review & Targeted Repair

OPEN ONLY AFTER a genuine saved attempt for the matching lesson. Compare reasoning, close Review, then do a changed retry.

DE08-01 - Ingestion architecture: source -> landing -> staging

Expected reasoning / normal control

Strong answer separates immutable raw evidence from parsed staging and later transformation. Repair if you overwrite source bytes or cannot say what one row/object means in staging.

Targeted repair

Identify the earliest failed boundary, write the defect in one sentence, repair only that competency, then close this PDF and repeat the task with changed input. Preserve the failed attempt as evidence; do not overwrite history.

DE08-02 - Defensive CSV ingestion

Expected reasoning / normal control

Expected normal fixture: 6 rows seen = 5 accepted + 1 quarantined because 2026-09-31 is invalid. Repair any implementation that silently drops the bad row.

Targeted repair

Identify the earliest failed boundary, write the defect in one sentence, repair only that competency, then close this PDF and repeat the task with changed input. Preserve the failed attempt as evidence; do not overwrite history.

DE08-03 - JSON and nested-record ingestion

Expected reasoning / normal control

Expected output grain is one row per tracking event; 5 events. Preserve raw nested JSON. Repair if you flatten shipment/device attributes but accidentally duplicate/drop events.

Targeted repair

Identify the earliest failed boundary, write the defect in one sentence, repair only that competency, then close this PDF and repeat the task with changed input. Preserve the failed attempt as evidence; do not overwrite history.

DE08-04 - Parquet ingestion and metadata inspection

Expected reasoning / normal control

Supplied fixture is 1736 bytes and starts/ends with PAR1. Footer length/offset can be read without PyArrow; schema/row groups require a compatible reader. Never invent metadata you did not inspect.

Targeted repair

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DE08-05 - REST API contracts, status codes and timeouts

Expected reasoning / normal control

Without auth /records returns 401. /slow with a 0.05s timeout must demonstrate timeout behavior. JSON body alone does not imply success.

Targeted repair

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DE08-06 - API authentication, pagination and rate limits

Expected reasoning / normal control

Expected successful fetch contains 7 unique IDs over pages 1-3. Protect against page loops and keep Bearer token out of committed production source/config.

Targeted repair

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DE08-07 - Retries, backoff, jitter and transient failures

Expected reasoning / normal control

The controlled server injects one 429 and one 503 before success. A bounded client should log two retry events and still retrieve 7 records. Repair unbounded retry or retry-on-401 logic.

Targeted repair

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DE08-08 - Database extraction and safe query boundaries

Expected reasoning / normal control

The controlled 10:00 inclusive / 12:30 exclusive extract returns 5 rows (O3-O7). Parameter placeholders must be used; no source mutation.

Targeted repair

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DE08-09 - Historical batch landing and raw metadata

Expected reasoning / normal control

Three batch files contain 9 business rows. Landing should preserve 3 distinct SHA-256 identities and identical bytes on rerun.

Targeted repair

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DE08-10 - Incremental loads and high-water marks

Expected reasoning / normal control

Expected runs: first 5, second 3, third 0. Composite cursor after first run is (2026-09-01T10:10:00Z, I005), so I006 at same timestamp is still eligible.

Targeted repair

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DE08-11 - CDC concepts and change-event semantics

Expected reasoning / normal control

Five CDC events produce current keys A1 and A3; A1 status PAID, A2 deleted. Repair code that treats delete as null-value update without a declared policy.

Targeted repair

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DE08-12 - Schema drift, compatibility, time zones and quarantine

Expected reasoning / normal control

Expected: 6 seen = 4 accepted + 2 quarantined. Valid +05:30 timestamps normalize to UTC. Repair coercion of bad amount/time to zero/current time.

Targeted repair

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DE08-13 - Checkpoint/state management and replay safety

Expected reasoning / normal control

Stored cursor must remain I005 after simulated destination failure. Advancing to I006 before commit creates a permanent skip risk.

Targeted repair

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DE08-14 - Idempotent landing and content identity

Expected reasoning / normal control

Normal batch has 3 unique content hashes; rerun keeps 3 landed files. Filename is not sufficient content identity; hash/business key policy must be explicit.

Targeted repair

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DE08-15 - Late-arriving data, event time and processing time

Expected reasoning / normal control

At 10-minute tolerance, L3 and L5 are late. Event time and arrived_at are different facts; neither should be overwritten by the other.

Targeted repair

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DE08-16 - Ingestion observability: counts, latency and run metadata

Expected reasoning / normal control

Run metadata must surface retries and quarantine counts. The normal smoke has 2 API retries; secrets must not appear in logs.

Targeted repair

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DE08-17 - Ingestion integration, reconciliation and backfill proof

Expected reasoning / normal control

Normal integration controls include batch rows 9, incremental rows 8, CDC current entities 2, and 3 quarantined rows across CSV+drift exercises. Rerun must not duplicate landed content.

Targeted repair

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